Causal Inference and Randomized Experiments

Lecture 6 (R code)

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Reading

This lecture covers material from Wooldridge (2025):

- Chapter 2.7 "Regression on a Binary Explanatory Variable"
  - Chapter 2.7a "Counterfactual (or Potential) Outcomes, Causality, and Policy Analysis"
- Chapter 7.1 "Describing Qualitative Information"
- Chapter 7.2 "A Single Dummy Independent Variable"
- Chapter 7.3 "Using Dummy Variables for Multiple Categories"
- Chapter 7.6b "Regression Adjustment under Randomized Assignment"

We will discuss the earlier part of Chapter 7.6 (7.6a "Program Evaluation and Unrestricted Regression Adjustment") in Week 8.

Chapter 7.1-7.3 should probably have been assigned much earlier. However, if you haven't read them, now is a good time to. The lecture this week will not discuss multiple categories as much, as we will focus on understanding potential outcome and causation.

Example: Dizon-Ross (2019)

Parents' Beliefs about Their Children's Academic Ability: Implications for Educational Investments[†]

By Rebecca Dizon-Ross[*]

i In a nutshell

Rural Malawi in 2012. School is free, but parents still spend money on their children's education. They have to decide on how much to invest based on what they know their children's academic potential. In reality, they have very poor (inaccurate) knowledge of their children's academic performance.

What would happen if you provided parents with more accessible information? Would they invest more? Differently?

Research Design

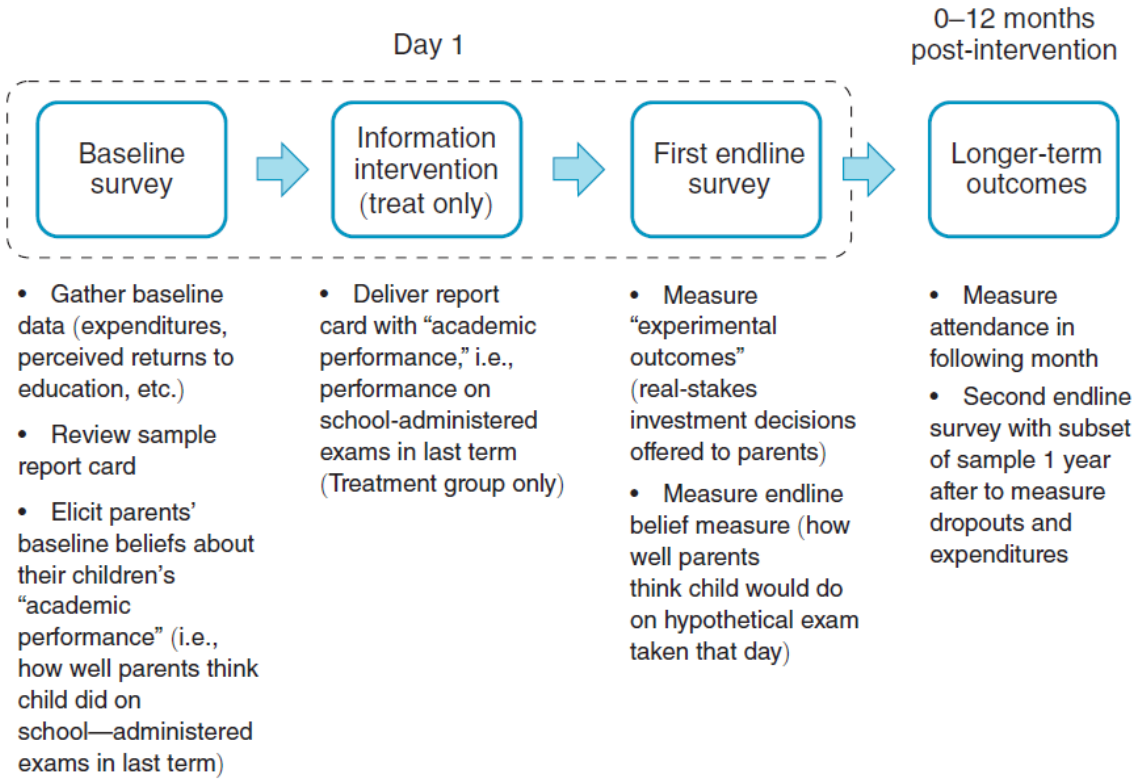


Figure 1: Figure 2 (p. 2741)

Balance

TABLE 1—BASELINE SUMMARY STATISTICS

	Full sample		Control	Treat	Treat – Control		
	Mean	SD	Mean	Mean	Mean	SE	*p*-val T=C
Panel A. Respondent background							
Female	0.77	0.42	0.77	0.76	−0.01	0.02	0.37
Primary education decision maker	0.92	0.27	0.91	0.92	0.01	0.01	0.31
Age	40.8	11.0	40.6	41.0	0.32	0.44	0.47
Education (years)	4.44	3.57	4.42	4.45	0.04	0.13	0.78
Respondent has secondary education+	0.11	0.31	0.11	0.11	0.01	0.01	0.62
Parent can read or write Chichewa	0.67	0.47	0.67	0.68	0.01	0.02	0.67
Respondent is farmer	0.46	0.5	0.47	0.46	−0.01	0.02	0.7
Respondent's weekly income	2,126	4,744	2,051	2,203	197	194	0.31

Figure 2: Table 1 (p. 2743)

Simple difference between means

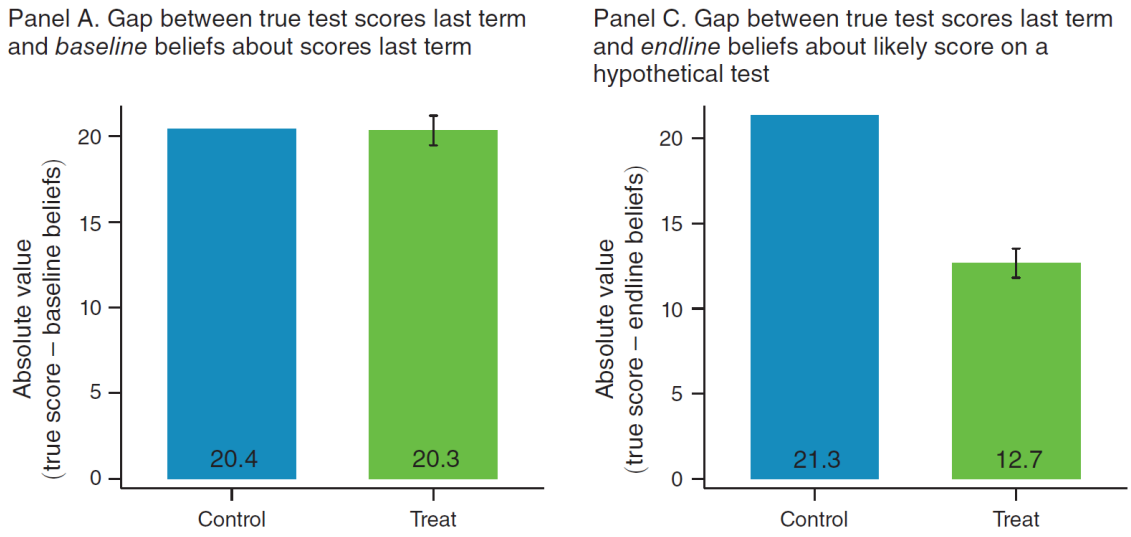


Figure 3: Figure 3 (p. 2745)

Heterogeneity

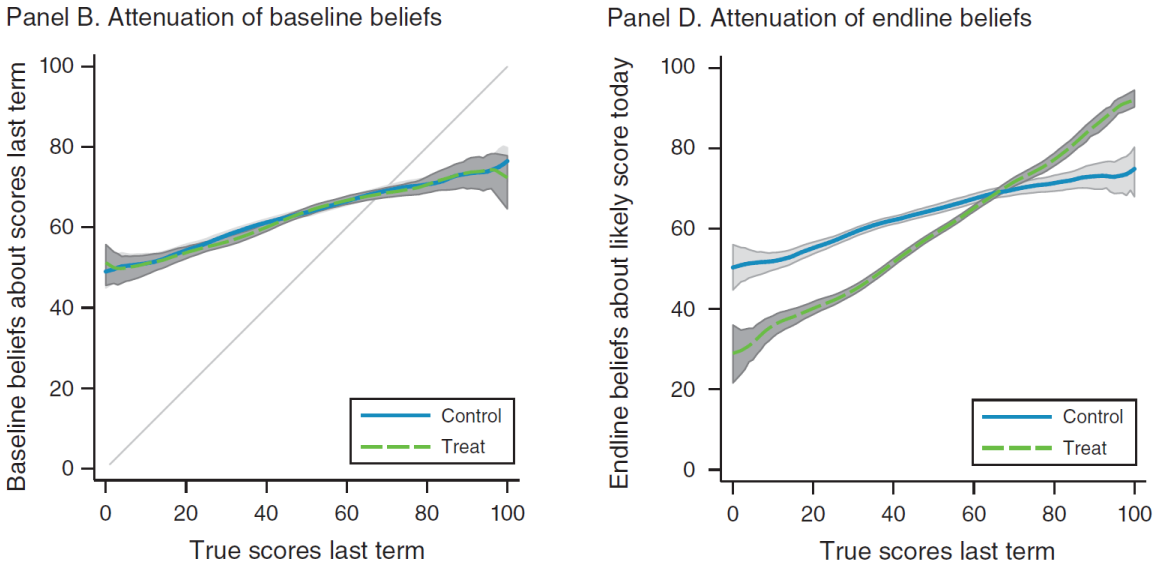


Figure 4: Figure 3 (p. 2745)

Regression

TABLE A.1—AVERAGE TREATMENT EFFECTS

	Experimental outcomes				Non-experimental outcomes		
	Endline beliefs	Math workbook difficulty level	English workbook difficulty level	ln(math textbook WTP) − ln(English textbook WTP)	Enrollment	ln(Total educ. expenditures)	Attendance rate
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A. Average treatment effects							
Treat	−7.47	−32.4	−13.7	0.14	−0.39	0.0024	−0.16
	[0.53]	[2.16]	[2.25]	[0.041]	[0.71]	[0.049]	[0.79]
Observations	5,244	5,239	5,239	5,219	1,786	1,709	1,827

Notes: Data sources are baseline survey, baseline test score data, both endline surveys, and endline administrative data. Each observation is a child. Standard errors are clustered at the household level. Regressions control for school FE, average parental years of education, parent gender, a parental edtucation proxy used for stratification, the between-child score gap, child baseline performance, child gender, grade FE, and the baseline value of the dependent variable (baseline value not available for enrollment or experimental outcomes). The dependent variable in Column 1 corresponds to the parent's endline beliefs about the child's overall score on a hypothetical test taken the same day as the endline survey. Workbook difficulty choices are coded as 0 for beginner, 100 for average, 200 for advanced. *Enrollment* is defined as being enrolled in school one year after the intervention; enrollment and attendance scaled to be out of 100 (so, for example, enrollment is equal to 100 if the child is still enrolled and 0 otherwise). Panel B uses the relevant measure of beliefs (e.g., overall for beliefs, math − English for textbooks; see Table 3 for details).

Potential Outcomes Framework

Neyman-Rubin Causal Model

The Potential Outcomes Framework is a key component of the Neyman-Rubin Causal Model[1]

- We won't cover all aspects of this model.
- I will reference texts in green boxes that you can visit to read further

> **On Causation**
>
> The Neyman-Rubin Causal Model is not the only framework for defining causal relationships. In addition, it is a model well suited to the analysis of experimental data.

[1] Also referred to as the Rubin Causal Model (originally by Holland 1986). The framework is attributed to both Jerzy Neyman and Donald Rubin; as well as Paul Holland (Heckman and Pinto 2022).

Another causal framework Directed Acyclic Graphs (DAGs) has become very influential in computer and data sciences. See Pearl and Mackenzie (2018)'s now famous book, *They Book of Why*, on the topic. G. W. Imbens (2020) provides an excellent discussion of DAG versus POF, and why POF is a more natural fit for economists (see also Heckman and Pinto 2022).
Finally, there are other notions of causal relations in Econometrics. For example, in time series there is the concept of Granger causation which relates more to temporal prediction and structural modelling thinks about causation outside of the "no causation without manipulation" notion of causation within the Neyman-Rubin model (Rubin 1975, 238). You might enjoy Hoover (2008)'s discussion on the topic.

Component 1

The Neyman-Rubin model has three important components (G. W. Imbens and Rubin 2015)

> **Potential Outcomes**: The outcomes corresponding to different levels of treatment or manipulation: "no causation without manipulation" (Rubin 1975, 238)

- Potential outcomes are *a priori* observable, but *a posteriori* only one outcome is observable.
- A variable can only be a "cause" if it is possible, at least in principle, to conceive of a hypothetical or physical intervention ("treatment") that manipulates it. We tend not to discuss the causal effect of demographic characteristics.[2]

Components 2

> **Multiple units**: Given the limit on observation, one must observe multiple units to infer a causal effect.

- Need to make a key stability assumption: Stable Unit Treatment Value Assumption (SUTVA).

SUTVA

The Stable Unit Treatment Value Assumption (SUTVA) is an overarching assumption for this entire Neyman-Rubin Causal Model

> "The potential outcomes for any unit do not vary with the treatments assigned to other units, and, for each unit, there are no different forms or versions

[2]The caveat here might be hypothetical resume studies; e.g. Bertrand and Mullainathan (2004) and Oreopoulos (2011).

of each treatment level, which lead to different potential outcomes." (G. W. Imbens and Rubin 2015, 10)

This assumption,

1. **Rules out interference between treatment units.** This rules out any spillover effects between 'treatment' and 'control'.
2. **Ensures that there are no hidden variations in the treatment level.** It is okay for there to be different levels of treatment, so long as they are explicit and well-defined *a priori*.

Component 3

Assignment mechanism: The function that determines the manipulation of units.

Assignment may be:

- **probabilistic**; e.g. randomized experiment
- **deterministic**; e.g. thresholds/cutoffs
- **conditional**; depend partly depend on observable characteristics ($\mathbf{x}_i$).
- **controlled** (by the researcher); e.g. randomized control trial
- **exogenous** (to the treatment unit); e.g. public policy or natural experiments[3]
- **known** (by the researcher); e.g. controlled or natural experiments
- **unknown**; e.g. observational study,

Propensity score

If you continue to study causal inference you learn about methodologies like propensity score matching. The propensity score is the probability of treatment conditional on a set of observable characteristics.

$$\rho(\mathbf{x_i}) = Pr(w_i = 1|\mathbf{x}_i)$$

This measure is key to many causal inference results. You can re-frame the question "do we know the assignment mechanism" to "do we know the functional form of the propensity score".

[3]Often referred to as quasi-experiment.

Example

Consider the example of Dizon-Ross (2019): there are two possible *states of the world*:

Treated:

- receive information

Researcher observes:

$$y(\text{info})$$

Untreated: (*control*)

- no information (status-quo)

Researcher observes:

$$y(\text{status-quo})$$

We call these **potential outcomes**.

Possible outcomes

For a discrete outcome like enrollment there are 4 possible outcomes:

1. Enroll only with information

$$y(\text{info}) = \text{enroll} \quad \text{and} \quad y(\text{status-quo}) = \text{don't enroll}$$

2. Enroll, regardless

$$y(\text{info}) = \text{enroll} \quad \text{and} \quad y(\text{status-quo}) = \text{enroll}$$

3. Don't enroll, regardless

$$y(\text{info}) = \text{don't enroll} \quad \text{and} \quad y(\text{status-quo}) = \text{don't enroll}$$

4. Enroll only without information

$$y(\text{info}) = \text{don't enroll} \qquad \text{and} \qquad y(\text{status-quo}) = \text{enroll}$$

Causal effect

According to the Neyman-Rubin Causal Model, in:

- scenarios (1) **and** (4) there has been a **causal effect**
- scenarios (2) **and** (3) there has been NO *causal effect*.

> **Important**
>
> Notice two things (G. W. Imbens and Rubin 2015, 6),
>
> 1. "First, the definition of the causal effect depends on the potential outcomes, but it does not depend on which outcome is actually observed."
>
> 2 "Second, the causal effect is the comparison of potential outcomes, for the same unit, at the same moment in time post-treatment. In particular, the causal effect is not defined in terms of comparisons of outcomes at different times."

Notation

Let $w_i \in \{0, 1\}$ be an indicator of an individuals *discrete* treatment status.[4]:

- $y_i(1)$ be the potential outcome when treated
- $y_i(0)$ be the potential outcome when not treated[5]

> **Continuous Treatment**
>
> We focus here on discrete treatments. It is straight forward to incorporate additional, separate treatments, but to model a continuous treatment you need to make additional assumptions about the functional form of the causal model. The only assumption we have made here is that of **additive treatment effects**: $y_i(1) = y_i(0) + \tau_i$.

[4] This notation of w_i as an indicator of an unit's treatment status is common in the literature on experiments and is used in one of the seminal books on the topic: G. W. Imbens and Rubin (2015)'s *Causal Inference in Statistics, Social, and Biomedical Sciences*. You will see it in Wooldridge (2025, sec. 7.6).

[5] You will see other notations in the literature. For example, the use of superscripts $\{y_i^1, y_i^0\}$ or subscripts $\{y_{i1}, y_{i0}\}$.

The model may in fact be multiplicative: $y_i(1) = y_i(0) \times \tau_i$. This distinction becomes important when you consider that many researchers use this framework to estimate treatment effects measured in log-transformed outcome variables.
One solution to the continuous treatment problem is to think about different treatment "dosages" (as in medicine). This approach discretizes the continuous treatment and returns the problem to the simple binary case.

Observed outcome

A priori, both potential outcomes could be realised, but **in the data you observe only one**:

$$y_i = \begin{cases} y_i(1) & \text{if } w_i = 1 \\ y_i(0) & \text{if } w_i = 0 \end{cases}$$

> "fundamental problem of causal inference" (Holland 1986, 947)

Which we can write this as a linear function:

$$y_i = y_i(0) + w_i(y_i(1) - y_i(0))$$

ITE

The individual (unit) treatment effect (ITE) is defined as,

$$\tau_i = y_i(1) - y_i(0)$$

However, *you cannot observe (identify) or estimate the ITE*

WHY?

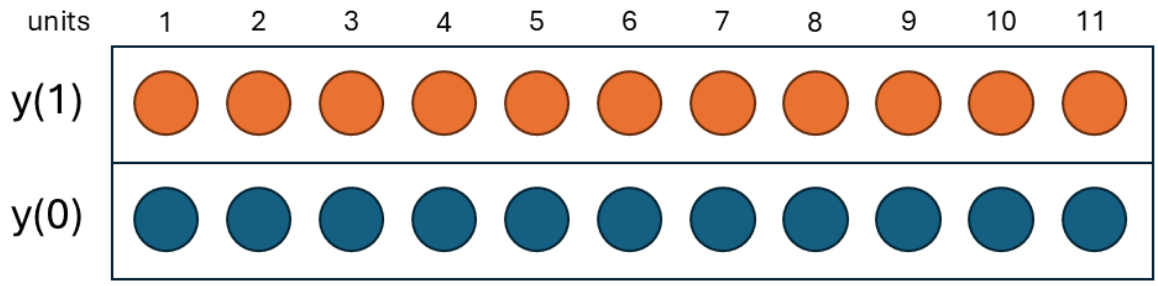
- it would require you to observe **both** potential outcomes of the same unit *in the same period*
- referred to as the "missing data problem" (G. W. Imbens and Rubin 2015, 34)

Notice!

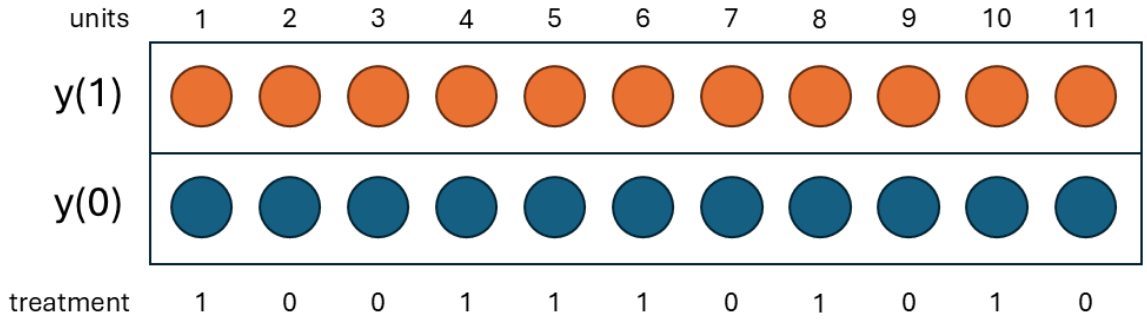
- the individual level treatment effect varies at the individual level: τ_i

Visualization

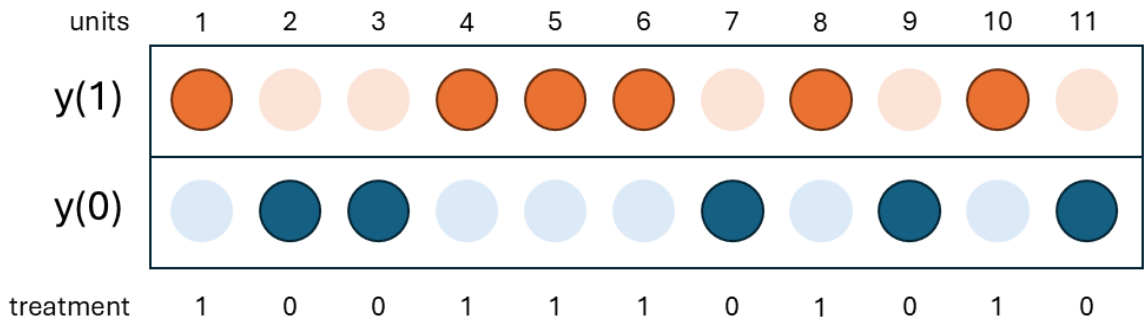
A priori, both potential outcomes are observable



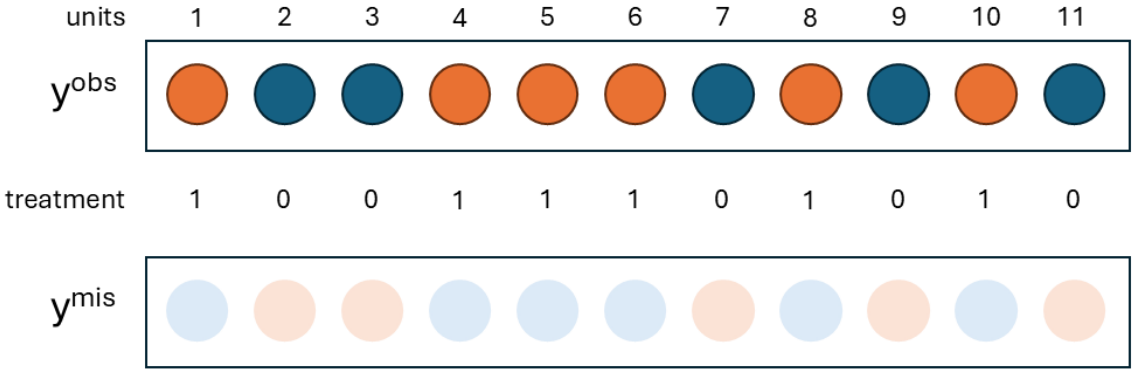
Then, treatment is assigned



Treatment determines which PO you observe



There is an observed outcome and (theoretically) a missing one



Average TEs

In place of the ITE, we focus on identifying and estimating average TEs

- Average Treatment Effect (ATE)

$$\tau_{ate} = E[\tau_i] = E[y_i(1) - y_i(0)]$$

- Average Treatment Effect of the Treated (ATT)

$$\tau_{att} = E[\tau_i | w_i = 1] = E[y_i(1) - y_i(0) | w_i = 1]$$

- Average Treatment Effect of the Untreated (ATU)

$$\tau_{atu} = E[\tau_i | w_i = 0] = E[y_i(1) - y_i(0) | w_i = 0]$$

These are called **causal estimands**

> Finite vs Super Population
>
> These are all super population casual estimands, denoted in terms of the expected values. There is a corresponding set of finite sample causal estimands in which the realized sample is held constant (see G. W. Imbens and Rubin 2015).
> In our discussion of regression models, the primary source of randomness was sampling. The super population definition of causal estimands takes the same approach: the expectation here is with respect to the (unobservable) population distributions.
> However, in this world of randomized experiments there is a second source of randomness: assignment (i.e. w_i). In a different realization of the experiment, a different set of individuals *within the same sample* could have been treated. You can therefore think about the ATE of the finite, fixed sample.
>
> $$\tau_{ate}^{fs} = \frac{1}{n}\sum_{i=1}^{n}\tau_i = \frac{1}{n}\sum_{i=1}^{n}(y_i(1) - y_i(0))$$
>
> As with the super population ATE, you can observe this in the data. It would require knowing both $y_i(1)$ and $y_i(0)$ for every individual.
> This finite sample approach tends not to be used extensively in applied economics research. Economists tend to take the approach of thinking about the fixed sample as a draw from the sampled (super) population and use their estimates to learn about the wider population. You see this in the way economists evaluate public policies.

ATE vs ATT

We prefer the **ATE** if,

- we want to learn about the effect on the entire population;
- i.e., mandatory exposure.
- or lottery

We prefer the **ATT** if,

- the population includes individuals that will not take up the treatment/policy;
- e.g., a policy that targets single mothers;

- or program is voluntary,

The ATT is "informative about the effect of entirely eliminating the current program" (G. W. Imbens and Wooldridge 2009, 14–15).

Assignment

Two groups

Suppose you treat one group of individuals, while observing another untreated. Then *in expectation*, you can learn about

$$E[y_i|w_i = 1] - E[y_i|w_i = 0]$$

Substituting in $y_i = y_i(0) + w_i(y_i(1) - y_i(0))$

$$E[y_i(1)|w_i = 1] - E[y_i(0)|w_i = 0]$$

Is this equal to either the ATE, ATT, or ATU? **NO!**

$$E[y_i(1)|w_i = 1] - E[y_i(0)|w_i = 0]$$

Why?

- The conditions are not the same!

Decomposition

Let us add and subtract:

$$E[y_i(0)|w_i = 1]$$

the **unobserved counterfactual** for the treated group.

$$E[y_i(1)|w_i = 1] - E[y_i(0)|w_i = 1] + E[y_i(0)|w_i = 1] - E[y_i(0)|w_i = 0]$$

The expression has two parts:

$$\begin{aligned} &\underbrace{E[y_i(1)|w_i = 1] - E[y_i(0)|w_i = 1]} + \underbrace{E[y_i(0)|w_i = 1] - E[y_i(0)|w_i = 0]} \\ = &\underbrace{E[y_i(1) - y_i(0)|w_i = 1]}_{\text{ATT}} + \underbrace{E[y_i(0)|w_i = 1] - E[y_i(0)|w_i = 0]}_{\text{selection}} \end{aligned}$$

> **!** Self-selection
>
> As alluded to earlier in Lecture 3, selection here is different to the problem of sample selection. This selection problem can more aptly described as the "self-selection problem" (Wooldridge 2025, 255).

Selection

The problem of selection **into treatment** is fundamental to any discussion of causation.

$$\underbrace{E[y_i(0)|w_i = 1] - E[y_i(0)|w_i = 0]}_{\text{selection}}$$

The term describes the difference between

1. an unobserved counterfactual outcome of the treated group: the outcome mean you *would have* observed, had the group NOT been treated

2. the observed outcome mean of the untreated (control) group

Unconfoundedness

In order to say much more, we need an assumption:

> **Unconfounded assignment:** assignment into the treatment is unconfounded if
>
> $$y_i(1), y_i(0) \perp w_i$$

And, **if assignment is random, then it is unconfounded**.

> **Note**
>
> Some people equate unconfoundedness with randomization, but they are distinct. Unconfoundedness is a property of an assignment mechanism: treatment assignment is independent of potential outcomes, possibly conditional on covariates. Randomization is one particular assignment mechanism that guarantees this property by design. In observational studies, by contrast, unconfoundedness can't be guaranteed and must instead be assumed.

> **Unconfounded vs exogenous**
>
> Unconfoundedness should also be distinguished from exogeneity, a term that is used in several different senses. In applied work it often means simply that a treatment is imposed from outside, as with a government policy change, but external imposition does not ensure that assignment is unrelated to the factors that determine outcomes. In econometrics it more formally means that a regressor is uncorrelated with the error term of a particular model, which is both a weaker condition than independence and one that depends on how the model is specified. Unconfoundedness, by contrast, is defined directly in terms of the assignment mechanism and the potential outcomes: assignment is independent of the potential outcomes given covariates, that is, as if randomly assigned.

Unconfounded assignment

1. $E[y_i(0)|w_i = 1] = E[y_i(0)|w_i = 0] = E[y_i(0)]$

$$\Rightarrow \underbrace{E[y_i(0)|w_i = 1] - E[y_i(0)|w_i = 0]}_{\text{selection}} = 0$$

2. $\underbrace{E[y_i(1) - y_i(0)|w_i = 1]}_{\text{ATT}} = \underbrace{E[y_i(1) - y_i(0)|w_i = 0]}_{\text{ATU}} = \underbrace{E[y_i(1) - y_i(0)}_{\text{ATE}}$

If assignment is unconfounded, then

$$E[y_i|w_i = 1] - E[y_i|w_i = 0] = \tau_{ate}$$

We therefore refer to unconfoundedness as the **identifying assumption**.

- without it we cannot learn about the ATE (or ATT) from the difference between two groups.

Testing unconfoundedness

In practice, researchers attempt to demonstrate that "randomization has worked" through the construction of a **balance table**

- Test difference in *baseline* covariates across treatment-control
- Can do a joint ANOVA test or adjust p-values for multiple inference.
- Covariates must be *observable* (hence, baseline survey)
- Researcher hopes to *fail to reject* H_0 : difference. Does not prove unconfounded assignment.

> **!** Expectation vs Realization
>
> Random assignment gives you unconfoundedness, which **in expectation** means that the treatment and control group are (theoretically) the same. The realized allocation will consist of two groups that are NOT the same.
> We can rely on LLN to say that for a large enough sample, the mean of each group approaches its expected value. But, many RCTs do not have very large samples.

Estimation

Estimator

Under unconfoundedness, the ATE has a simple difference between means (SDM) estimator (Wooldridge 2025, sec. 2.7a)

$$\hat{\tau} = \bar{y}_1 - \bar{y}_0$$

where,

$$\bar{y}_1 = \frac{1}{n_1} \sum_{i:w_i=1} y_i \qquad \text{and} \qquad \bar{y}_0 = \frac{1}{n_0} \sum_{i:w_i=0} y_i$$

and n_1 (n_0) is number of treated (control) units

Variance

The variance of this estimator is

$$Var(\hat{\tau}) = \frac{\sigma_0^2}{n_0} + \frac{\sigma_1^2}{n_1}$$

Which can be estimated by,

$$\widehat{Var}(\hat{\tau}) = \frac{\hat{\sigma}_0^2}{n_0} + \frac{\hat{\sigma}_1^2}{n_1}$$

where $\hat{\sigma}_j^2 = \frac{1}{n_j - 1} \sum_{i:w_i=j} (y_i - \bar{y}_j)^2$ for $j = 0, 1$.

- We call this the Neyman variance estimator within Causal Inference literature.[6]

[6] Also referred to as the Welch test/estimator.

Homogeneous vs heterogenous TEs

If TEs are **homogeneous**:

$$\tau_i = y_i(1) - y_i(0) = \tau \; \forall \; i$$

Then,

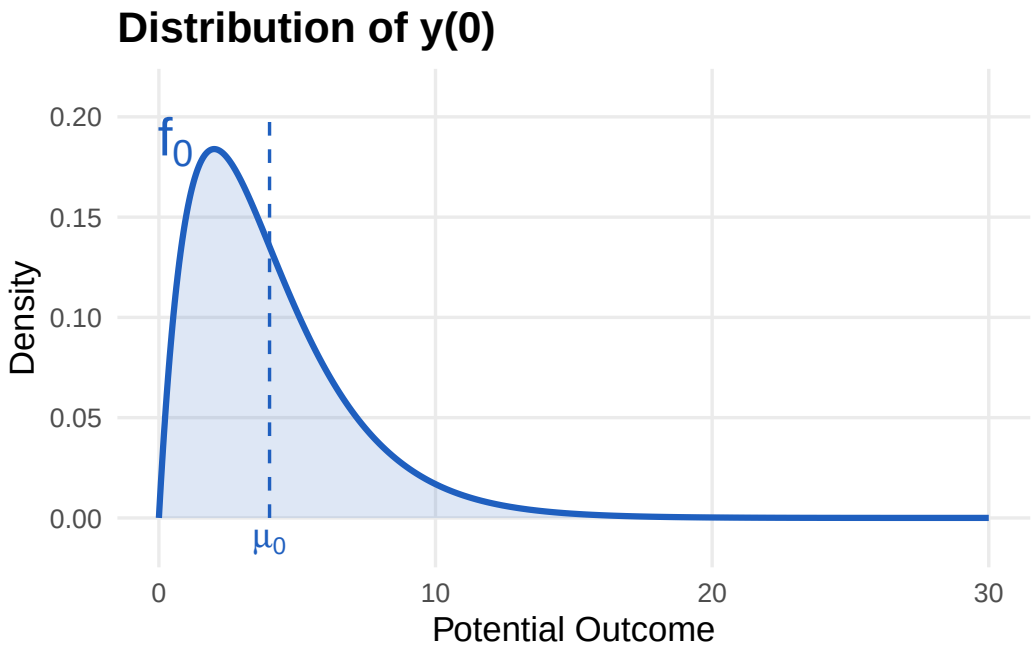
$$y_i = y_i(0) + w_i(y_i(1) - y_i(0)) = y_i(0) + w_i\tau$$

And,

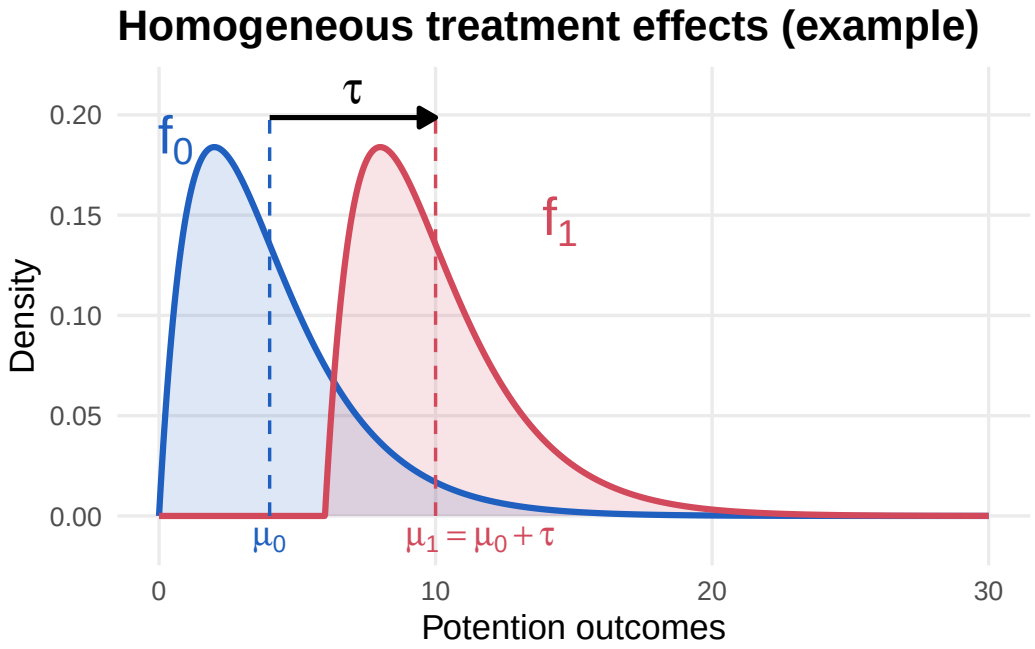
$$\sigma_0^2 = \sigma_1^2 = \sigma^2$$

Demonstration

Consider the following (simulated) distribution of y_0, which need not not be normal.



If TEs are homogeneous, then $y_i(1) = y_i(0) + \tau$ (i.e. a mean shift)



The variance does not change.

Homogenous TEs

If TEs are **homogenous**

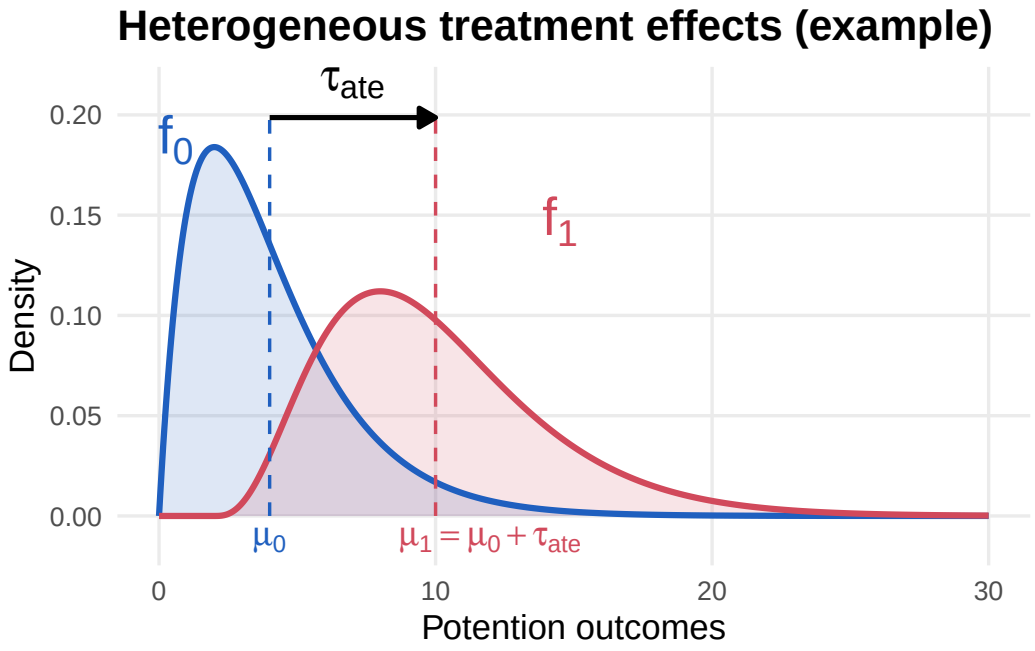
$$Var(\hat{\tau}) = \frac{\sigma_0^2}{n_0} + \frac{\sigma_1^2}{n_1} = \sigma^2\left(\frac{1}{n_1} + \frac{1}{n_0}\right)$$

Which can be estimated as,

$$\hat{\sigma}^2 = \frac{1}{n-2}((n_0 - 1) \times \hat{\sigma}_0^2 + (n_1 - 1) \times \hat{\sigma}_1^2)$$

We call this the **constant variance estimator**.

If the TEs are heterogeneous, then $y_i(1) = y_i(0) + \tau_i$



Distribution

Note, we do not know the distribution of $y_i(0)$ or $y_i(1)$!

- In applied work, researchers prefer not to make any assumptions
- Rely on CLT to show asymptotic distribution is normal
- In practice, and since RCTs often have a small sample, use t-distribution to be more conservative
- There are other approaches to inference (e.g., permutation tests like Fisher's exact p-value) that we won't cover here, but are more relevant for small samples.

In Stata/R

Example uses a 10% sample of the replication data for Dizon-Ross (2019).

Constant variance test

```r
t.test(belief_err_bl ~ treat, data = dr19_dta, var.equal = TRUE)
```

```
    Two Sample t-test

data:  belief_err_bl by treat
t = 0.47476, df = 1310, p-value = 0.635
alternative hypothesis: true difference in means between group 0 and group 1 is not equal to
95 percent confidence interval:
 -1.196795  1.960997
sample estimates:
mean in group 0 mean in group 1
       20.61721        20.23511
```

Unequal variance test (Neyman)

```r
t.test(belief_err_bl ~ treat, data = dr19_dta, var.equal = FALSE)
```

```
    Welch Two Sample t-test

data:  belief_err_bl by treat
t = 0.47432, df = 1299.8, p-value = 0.6354
alternative hypothesis: true difference in means between group 0 and group 1 is not equal to
95 percent confidence interval:
 -1.198271  1.962473
sample estimates:
mean in group 0 mean in group 1
       20.61721        20.23511
```

Linear Model

Dummy regressor

We know that for a discrete $x_i \in \{0, 1\}$,

$$y_i = \beta_0 + \beta_1 x_i + u_i$$

where,

$$\beta_1 = E[y_i|x_i = 1] - E[y_i|x_i = 0]$$

OLS

Moreover, we showed in Lecture 2 that the OLS estimator for β_1 is just,

$$\hat{\beta}_1 = \bar{y}_1 - \bar{y}_0$$

Translation of POF

We can translate the POF to a linear regression model,[7]

$$\begin{aligned}
y_i =& y_i(0) + w_i(y_i(1) - y_i(0)) \\
=& y_i(0) + \tau_i w_i \\
=& E[y_i(0)] + \tau_i w_i + y_i(0) - E[y_i(0)] \\
=& E[y_i(0)] + \tau_{ate} w_i + \underbrace{y_i(0) - E[y_i(0)] + (\tau_i - \tau_{ate}) w_i}_{u_i} \\
=& \alpha_0 + \tau_{ate} w_i + u_i
\end{aligned}$$

Consider the error term:

$$u_i = \underbrace{y_i(0) - E[y_i(0)]}_{1.} + \underbrace{(\tau_i - \tau_{ate})}_{2.} \underbrace{w_i}_{3.}$$

it includes,

1. the deviation of $y_i(0)$ from its unconditional mean
2. the devation of τ_i from its unconditional mean (i.e. τ_{ate}) $\Leftarrow$ **heterogeneity**
3. w_i is found in the error term, as well

[7] Wooldridge (2025, sec. 2.7a) also includes a version with homogeneous TEs. This is the more general case.

Wooldridge's notation

Wooldridge (2025, sec. 7.2a) defines:

$$\begin{aligned} u_i(0) =& y_i(0) - E[y_i(0)] \\ u_i(1) =& y_i(1) - E[y_i(1)] \end{aligned}$$

which then gives you,

$$u_i = u_i(0) = w_i(u_i(1) - u_i(0))$$

from p. 55. We will not use this notation here. However, it helps you, feel free to use it.

Solve for $E[u_i|w_i = 0]$

$$\begin{aligned} E[u_i|w_i = 0] =& E[y_i(0) - E[y_i(0)] + (\tau_i - \tau_{ate})w_i|w_i = 0] \\ =& E[y_i(0) - E[y_i(0)]|w_i = 0] \\ =& E[y_i(0)|w_i = 0] - E[y_i(0)] \end{aligned}$$

If assignment is unconfounded:

$$E[y_i(0)|w_i = 0] - E[y_i(0)] = E[y_i(0)] - E[y_i(0)] = 0$$

Solve for $E[u_i|w_i = 1]$

$$\begin{aligned} E[u_i|w_i = 1] =& E[y_i(0) - E[y_i(0)] + (\tau_i - \tau_{ate})w_i|w_i = 1] \\ =& E[y_i(0) - E[y_i(0)] + y_i(1) - y_i(0) - E[y_i(1)] + E[y_i(0)]|w_i = 1] \\ =& E[y_i(1) - E[y_i(1)]|w_i = 1] \\ =& E[y_i(1)|w_i = 1] - E[y_i(1)] \end{aligned}$$

Again, *if assignment is unconfounded*:

$$E[y_i(1)|w_i = 1] - E[y_i(1)] = E[y_i(1)] - E[y_i(1)] = 0$$

Thus, with unconfounded assignment we can write,

$$y_i = \alpha_0 + \tau_{ate} w_i + u_i$$

where,

- slope parameter on w_i identifies the ATE
- the error term fulfills MLR 4
- OLS estimator is unbiased (for the ATE)

> **Warning**
>
> If given the model $y_i = \beta_0 + \beta_1 x_i + u_i$ with $x_i = \{0, 1\}$, then it is trivial to say that $E[u_i|x_i] = 0$. This is because we can write the CEF of y given x as a linear function:
>
> $$E[y_i|x_i] = \underbrace{E[y_i|x_i = 0]}_{\beta_0} + \underbrace{(E[y_i|x_i = 1] - E[y_i|x_i = 0])}_{\beta_1} x_i$$
>
> All this means is that we can identify the mean of two groups. It does NOT mean that $\beta_1 = E[y_i|x_i = 1] - E[y_i|x_i = 0]$ is equal to the ATE. **For that, you need unconfoundedness.**
> Consider the case where $x_i = \mathbf{1}\{\text{“female”}\}$. Then β_1 tells you the mean difference between female and male observations. Would you describe this as the ATE of being female?

Error variance

We have shown that under unconfoundedness $E[u_i|w_i] = 0$.[8] What about:

$$Var(u_i|w_i) = E[u_i^2|w_i] = ?$$

- Suppose TEs are **homogenous**: $\tau_i = \tau = \tau_{ate} \; \forall \; i$

$$u_i = y_i(0) - E[y_i(0)] + \underbrace{(\tau_i - \tau_{ate})}_{=0} w_i = y_i(0) - E[y_i(0)]$$

[8] Where $u_i = y_i - \alpha_0 - \tau_{ate} w_i$.

Homoskedasticity

If treatment effects are **homogeneous**, then

$$E[u_i^2|w_i] = E[(y_i(0) - E[y_i(0)])^2|w_i] = E[(y_i(0) - E[y_i(0)])^2] = \sigma_0^2 = \sigma^2$$

under unconfoundedness.

- homogeneity ⇒ homoskedastic error variance

Heteroskedasticity

If TEs are **heterogenous**:

$$E[u_i^2|w_i] = \begin{cases} E[(y_i(0) - E[y_i(0)])^2] = \sigma_0^2 & \text{if } w_i = 0 \\ E[(y_i(1) - E[y_i(1)])^2] = \sigma_1^2 & \text{if } w_i = 1 \end{cases}$$

under unconfoundedness.

- heterogeneity ⇒ heteroskedastic error variance

Deaton's critique

Angus Deaton - a noble laureate in economics - is famous for not being overly enthusiastic about the use of RCTs in Economics; in particular, in Development Econonomics (see Athey and Imbens 2017). His critiques of the field range from the ethics of RCTs in poor countries, questions about the real-world policy relevance of RCT-based causal estimates, and more technical points G. Imbens (2018).

> ""I argue that evidence from randomized experiments has no special priority. … Randomized experiments cannot automatically trump other evidence, they do not occupy any special place in some hierarchy of evidence" (Deaton 2010, 426)

One technical point Deaton emphasized - especially early on - was the apparent lack of attention/care shown to the correct estimation of SEs in RCT-based research.

> "If the regression … is run with the standard heteroskedasticity correction to the standard error, the result will be the same as the formula for the standard error of the difference between two means, but not otherwise except in the special case where there are equal numbers of experimental and controls, in which case it turns out that the correction makes no difference and the OLS standard error is correct. It is not clear in the experimental development

literature whether the correction is routinely done in practice…" (Deaton 2010, 442)

It is fair to say that the research on best practice has come a long way since then and there is a vaste literature topics like SE estimation for randomized experiments (Athey and Imbens 2017). Moreover, the debate over the significance of RCTs in Economics has largely been won, and not by Deaton. In 2019 Esther Duflo, Abhijit Banerjee, and Michael Kremer won the Nobel Prize in Economics "for their experimental approach to alleviating global poverty". The use of RCTs to answer important empirical questions about how the lives of the poor has fundamentally shaped how we design and evaluate real world policies (see Duflo and Banerjee 2011).

Variance estimator

We know from Lecture 3 that the (conditional) variance of the OLS estimator for τ_{ate} is,

$$\widehat{Var}(\hat{\tau}_{ols}|W) = \frac{\hat{\sigma}^2_{ols}}{SST_w}$$

under MLR 5 ($\Leftarrow$ homogeneous TEs). Where,

$$\hat{\sigma}^2_{ols} = \frac{1}{n-2}\sum_{i=1}^{n}\hat{u}_i^2$$

This is mathematically equivalent to the constant variance estimator.

$$= \frac{1}{n-2}((n_0-1)\times\hat{\sigma}^2_0 + (n_1-1)\times\hat{\sigma}^2_1)$$

i Proof

The proof follows a simple trick of splitting the sample into two groups: (1) $w_i = 0$ and (2) $w_i = 1$.

$$\begin{aligned}
\sum_{i=1}^{n}\hat{u}_i^2 &= \sum_{i=1}^{n}(y_i - \hat{\alpha}_0 - \hat{\tau}_{ols}w_i)^2 \\
&= \sum_{i:w_i=0}(y_i - \hat{\alpha}_0 - \hat{\tau}_{ols}w_i)^2 + \sum_{i:w_i=1}(y_i - \hat{\alpha}_0 - \hat{\tau}_{ols}w_i)^2 \\
&= \sum_{i:w_i=0}(y_i - \hat{\alpha}_0)^2 + \sum_{i:w_i=1}(y_i - \hat{\alpha}_0 - \hat{\tau}_{ols})^2
\end{aligned}$$

Since $\hat{\alpha}_0 = \bar{y}_0$ and $\hat{\alpha}_0 + \hat{\tau}_{ols} = \bar{y}_1$, the above is just

$$\begin{aligned}&\sum_{i:w_i=1}(y_i - \bar{y}_0)^2 + \sum_{i:w_i=0}(y_i - \bar{y}_1)^2 \\ =&(n_0 - 1) \times \hat{\sigma}_0^2 + (n_1 - 1) \times \hat{\sigma}_1^2\end{aligned}$$

In the denominator is

$$SST_w = n\bar{w}(1 - \bar{w})$$

This formula is quadratic in $\bar{w}$ - the share of the sample treated.

- At $\bar{w} = 0.5$, SST_w is maximized and $Var(\hat{\tau}^{OLS}|W)$ is minimized (holding other factors fixed).
- A balanced treatment is good for statistical power, as is a larger sample size.

Proof

This proof follows the same simple trick of splitting the sample into two groups: (1) $w_i = 0$ and (2) $w_i = 1$.

$$\begin{aligned}SST_w =& \sum_{i=1}^{n}(w_i - \bar{w})^2 \\ =& \sum_{i:w_i=0}(0 - \bar{w})^2 + \sum_{i:w_i=1}(1 - \bar{w})^2 \\ =& n_0\bar{w}^2 + n_1(1 - \bar{w})^2 \\ =& \underbrace{n(1 - \bar{w})}_{=n_0}\bar{w}^2 + \underbrace{n\bar{w}}_{=n_1}(1 - \bar{w})^2 \\ =& n\bar{w}(1 - \bar{w})(\bar{w} + 1 - \bar{w}) \\ =& n\bar{w}(1 - \bar{w})\end{aligned}$$

The proof exploits the fact that $\bar{w}$ is just the share of the sample that is treated: n_1/n. Likewise, $1 - \bar{w} = n_0/n$, the share of the sample untreated.

What if MLR 5 ($\Leftarrow$ heterogeneous TEs) does not hold?

- The White (1980) heteroskedastic-consistent estimator from Lecture 5 was,

$$HC0 = \frac{\sum_{i=1}^{n}(w_i - \bar{w})^2 \hat{u}_i^2}{\text{SST}_w^2}$$

It can be shown (we will not) that,

$$HC0 \neq \frac{\hat{\sigma}_0^2}{n_0} + \frac{\hat{\sigma}_1^2}{n_1} = HC2$$

The Neyman estimator is numerically equivalent to HC2.

- in practice, all HC0-HC3 estimators are asymptotically consistent.
- HC3 is recommended for smaller samples.

Proof

You can use the same trick of splitting the summation to show,

$$\begin{aligned}
\widehat{Var}(\hat{\tau}_{ols}|W) =& \frac{\sum_{i=1}^{n}(w_i - \bar{w})^2 \hat{u}_i^2}{\text{SST}_w^2} \\
=& \frac{\sum_{i:w_i=1}^{n}(1-\bar{w})^2(y_i - \hat{\alpha}_{ols} - \hat{\tau}_{ols})^2 + \sum_{i:w_i=0}^{n}(-\bar{w})^2(y_i - \hat{\alpha}_{ols})^2}{\text{SST}_w^2} \\
=& \frac{(1-\bar{w})^2 \sum_{i:w_i=1}^{n}(y_i - \hat{\alpha}_{ols} - \hat{\tau}_{ols})^2 + \bar{w}^2 \sum_{i:w_i=0}^{n}(y_i - \hat{\alpha}_{ols})^2}{\text{SST}_w^2} \\
=& \frac{(1-\bar{w})^2(n_1-1)\hat{\sigma}_1^2 + \bar{w}^2(n_0-1)\hat{\sigma}_0^2}{(n\bar{w}(1-\bar{w}))^2} \\
=& \frac{n_1-1}{n_1^2}\hat{\sigma}_1^2 + \frac{n_0-1}{n_0^2}\hat{\sigma}_0^2
\end{aligned}$$

And HC1 (Stata's robust default) will be,

$$\frac{n}{n-2} \times \left[\frac{n_1-1}{n_1^2}\hat{\sigma}_1^2 + \frac{n_0-1}{n_0^2}\hat{\sigma}_0^2\right]$$

It is evident that for n large, these estimators will be very similar to Neyman's estimator. You can show that HC2 is algebraically equivalent to Neyman (also referred to as Welch test/estimator).

Regression Adjustment

Purpose

The variance of the estimator is (assuming homogeneous TEs),

$$Var(\hat{\tau}_{ols}|W) = \frac{Var(u_i|w_i)}{SST_w} = \frac{\sigma_u^2}{SST_w}$$

where u_i is from the model,

$$y_i = \alpha_0 + \tau_{ate} w_i + u_i$$

- The only variation in y explained by this model is the difference between treatment-control.
- Will result in high variance and low-powered test.

Covariates

Suppose,

$$u_i = \gamma x_i + v_i$$

where x is a regressor measured before assignment[9]

- likely explain part of the outcome (i.e. $\gamma \neq 0$)
- commonly referred to as baseline covariates in RCTs

Regression adjustment

Instead of the simple regression, we estimate

$$y_i = \alpha_0 + \tau_{ate} w_i + \gamma x_i + v_i$$

- Wooldridge (2025, 257) refers to this as the Restricted Regression Adjustment (RRA)

[9]We can extend the notation to include multiple regressors, but it makes some of the math a little difficult. The conclusion does not change.

Variance

From Lecture 3 we know,

$$Var(\hat{\tau}_{rra}|W,X) = \frac{Var(v_i|w_i,x_i)}{SSR_w} = \frac{\sigma_v^2}{SSR_w}$$

where,

$$SSR_w = SST_w(1-R_w^2)$$

- R_w^2 is the amount of variance in assignment explained by the covariates. **This should be low, but won't be zero.**
- While $R_w^2 \neq 0$ in the finite sample, $R_w^2 \to 0$ as $n \to \infty$
- So, $SSR_w \approx SST_w$.

Error variance

The improvement in power comes from the fact that,

$$\underbrace{Var(u_i|w_i)}_{\sigma_u^2} = Var(\gamma x_i + v_i|w_i) = \underbrace{\gamma^2 Var(x_i)}_{>0} + \underbrace{Var(v_i|w_i,x_i)}_{\sigma_v^2}$$

assuming

- $Cov(x_i, v_i|w_i) = 0$
- w_i is assigned independently of x_i: $Var(x_i|w_i) = Var(x_i)$
- $Var(v_i|w_i) = Var(v_i|w_i,x_i)$ ($\Leftarrow$ heterogeneity unrelated to x)

Power

Putting these two together,

$$Avar(\hat{\tau}_{rra}) \leq Avar(\hat{\tau}_{ols})$$

where $Avar()$ is the asymptotic variance.[10]

Adding covariates,

- Reduces variance of the estimator by reducing the error variance
- Little trade-off in terms of the correlation between w and x (i.e., R^2_w) under assignment that is unconfounded (w.r.t. x's).
  - Any finite sample correlation disappears as $n \to \infty$.
  - **Covariates must not be potential outcomes of the experiment** (i.e. baseline).

Heterogeneity

This simple story is somewhat complicated by heterogeneity:

- heterogeneity is sits in the error term
- heterogeneity is likely related to x
- heterogeneity (that depends on x) introduces small-sample bias in models of the form:

$$y_i = \alpha_0 + \tau_{ate} w_i + \gamma x_i + v_i$$

- OLS remains consistent (for the ATE), despite this small-sample bias (Freedman 2008)

[10] This is a place holder for $p\lim \hat{\sigma}^2/(n^{-1}SSR_w)$.

URA

To avoid this small-sample bias, you can do Unrestricted Regression Adjustment (URA) (Wooldridge 2025, 257, 259; Negi and Wooldridge 2021).

$$y_i = \alpha_0 + \tau_{ate} w_i + \gamma x_i + \eta w_i \times (x_i - \bar{x}) + \varepsilon_i$$

- includes an interaction between the treatment indicator and the covariate(s)
  - models the heterogeneity related x
- the interactions include demeaned covariate(s)
  - this changes the interpretation of η (see Wooldridge 2025, sec. 6.2d)
  - ensures that the coefficient on w remains the ATE

Example: Reivisiting Dizon-Ross (2019)

RCT

Why randomize?

- Resolves the problem of self-selection
- Information parents have is endogenous
- Randomizing who gets information resolves this (in expectation)

Parents' Beliefs about Their Children's Academic Ability: Implications for Educational Investments[†]

By Rebecca Dizon-Ross[*]

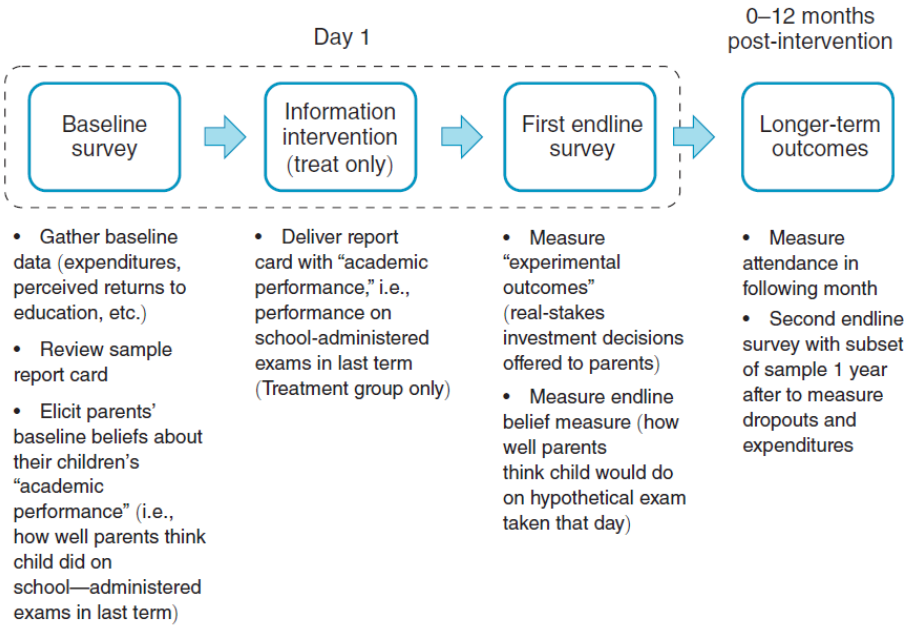
Research Design

Baseline survey:

- Measure covariates: balance and power

Endline survey:

- Measure outcomes
- Need to think about attrition



Balance

Balance:

- Test on observables
- Doesn't consider unobservables
- Will be non-zero correlation in data
- Validates research design

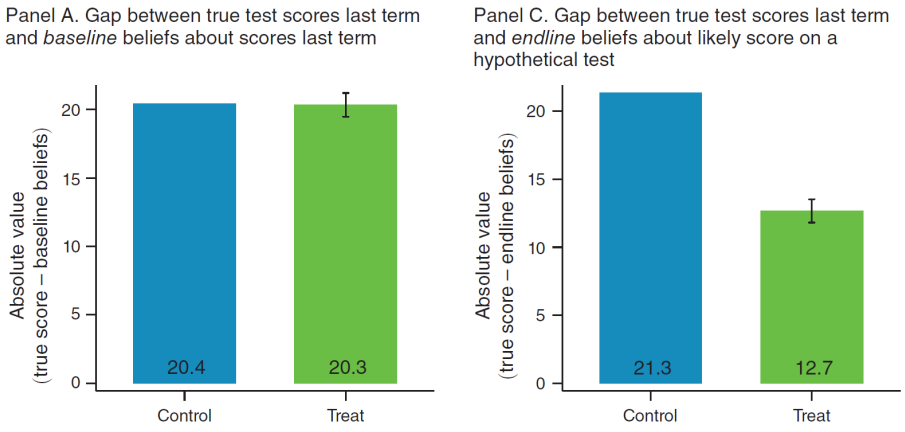
TABLE 1—BASELINE SUMMARY STATISTICS

	Full sample		Control	Treat	Treat – Control		
	Mean	SD	Mean	Mean	Mean	SE	*p*-val T=C
Panel A. Respondent background							
Female	0.77	0.42	0.77	0.76	−0.01	0.02	0.37
Primary education decision maker	0.92	0.27	0.91	0.92	0.01	0.01	0.31
Age	40.8	11.0	40.6	41.0	0.32	0.44	0.47
Education (years)	4.44	3.57	4.42	4.45	0.04	0.13	0.78
Respondent has secondary education+	0.11	0.31	0.11	0.11	0.01	0.01	0.62
Parent can read or write Chichewa	0.67	0.47	0.67	0.68	0.01	0.02	0.67
Respondent is farmer	0.46	0.5	0.47	0.46	−0.01	0.02	0.7
Respondent's weekly income	2,126	4,744	2,051	2,203	197	194	0.31

Simple difference between means

SDM:

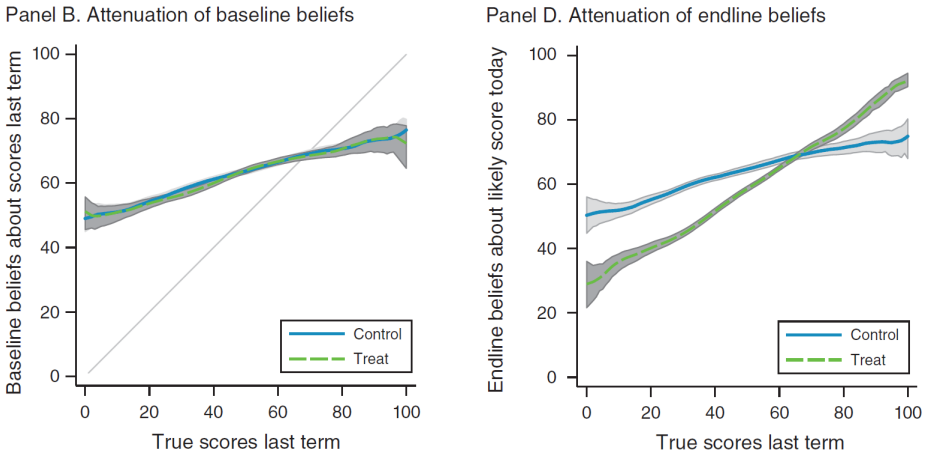
- Measures ATE under unconfoundedness
- Might have a high variance



Heterogeneity

Heterogeneity:

- POF developed to understand heterogeneous TEs
- Researchers will often commit to important margins of heterogeneity in their pre-analysis.



Regression

Regression adjustment:

- Can correct for any imbalance
- More valuable as a source of improved power
- Covariates from baseline (i.e. not outcomes of experiment)

TABLE A.1—AVERAGE TREATMENT EFFECTS

	Experimental outcomes				Non-experimental outcomes		
	Endline beliefs	Math workbook difficulty level	English workbook difficulty level	ln(math textbook WTP) − ln(English textbook WTP)	Enrollment	ln(Total educ. expenditures)	Attendance rate
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A. Average treatment effects							
Treat	−7.47	−32.4	−13.7	0.14	−0.39	0.0024	−0.16
	[0.53]	[2.16]	[2.25]	[0.041]	[0.71]	[0.049]	[0.79]
Observations	5,244	5,239	5,239	5,219	1,786	1,709	1,827

References

Bibliography

Athey, Susan, and Guido W Imbens. 2017. "The Econometrics of Randomized Experiments." In *Handbook of Economic Field Experiments*, 1:73–140. Elsevier.

Bertrand, Marianne, and Sendhil Mullainathan. 2004. "Are Emily and Greg More Employable Than Lakisha and Jamal? A Field Experiment on Labor Market Discrimination." *American Economic Review* 94 (4): 991–1013.

Deaton, Angus. 2010. "Instruments, Randomization, and Learning about Development." *Journal of Economic Literature* 48 (2): 424–55.

Deaton, Angus, and Nancy Cartwright. 2018. "Understanding and Misunderstanding Randomized Controlled Trials." *Social Science & Medicine* 210: 2–21.

Dizon-Ross, Rebecca. 2019. "Parents' Beliefs about Their Children's Academic Ability: Implications for Educational Investments." *American Economic Review* 109 (8): 2728–65.

Duflo, Esther, and Abhijit Banerjee. 2011. *Poor Economics*. Vol. 619. PublicAffairs, New York.

Freedman, David A. 2008. "On Regression Adjustments to Experimental Data." *Advances in Applied Mathematics* 40 (2): 180–93.

Heckman, James J, and Rodrigo Pinto. 2022. "Causality and Econometrics." National Bureau of Economic Research.

Holland, Paul W. 1986. "Statistics and Causal Inference." *Journal of the American Statistical Association* 81 (396): 945–60.

Hoover, Kevin D. 2008. "Causality in Economics and Econometrics." In *The New Palgrave Dictionary of Economics*, 1–13. Springer.

Imbens, Guido. 2018. "Understanding and Misunderstanding Randomized Controlled Trials: A Commentary on Deaton and Cartwright."

Imbens, Guido W. 2020. "Potential Outcome and Directed Acyclic Graph Approaches to Causality: Relevance for Empirical Practice in Economics." *Journal of Economic Literature* 58 (4): 1129–79.

Imbens, Guido W, and Donald B Rubin. 2015. "Causal Inference in Statistics, Social, and Biomedical Sciences." *New York* 517.

Imbens, Guido W, and Jeffrey M Wooldridge. 2009. "Recent Developments in the Econometrics of Program Evaluation." *Journal of Economic Literature* 47 (1): 5–86.

Negi, Akanksha, and Jeffrey M Wooldridge. 2021. "Revisiting Regression Adjustment in Experiments with Heterogeneous Treatment Effects." *Econometric Reviews* 40 (5): 504–34.

Oreopoulos, Philip. 2011. "Why Do Skilled Immigrants Struggle in the Labor Market? A Field Experiment with Thirteen Thousand Resumes." *American Economic Journal: Economic Policy* 3 (4): 148–71.

Pearl, Judea, and Dana Mackenzie. 2018. "The Book of Why."

Rubin, Donald B. 1975. "Bayesian Inference for Causality: The Importance of Randomization." In *Proceedings of the Social Statistics Section of the American Statistical Association*, 233–39. American Statistical Association, Alexandria, VA.

Wooldridge, Jeffrey M. 2025. "Introductory Econometrics: A Modern Approach."